Status	Finished
Started	Monday, 26 October 2026, 6:47 PM
Completed	Monday, 26 October 2026, 6:50 PM
Duration	3 mins 17 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Insertion Sort
- b. Quick Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. arrange()
- c. sort()
- d. sorted()

The correct answer is: sort()

Question **3**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is less efficient than bubble sort
c. It is based on the divide-and-conquer approach
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **4**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are split again into smaller sublists
c. They are compared element by element
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **5**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. The parts are merged without sorting
c. The parts are ignored
d. Each part is sorted independently

The correct answer is: Each part is sorted independently

Question **6**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 40,80,90,150,200,300,400,470,70,120
- c. 200,470,80,150,40,90,300,400,70,120
- d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **7**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Arranging data in a particular format
- c. Inserting data into a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Bubble Sort
- b. Insertion Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Quick sort
c. Selection sort
d. Merge sort

The correct answer is:
Merge sort

Question **10**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By comparing elements sequentially
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **11**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the largest element to its correct position
b. Dividing the list into halves
c. Sorting the entire list in one pass
d. Bubbling up the smallest element

The correct answer is: Bubbling up the largest element to its correct position

Question **12**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Greedy method
- c. Dynamic programming
- d. Backtracking

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Complexity
- c. Merging
- d. Sorting

The correct answer is:
Complexity

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data sequentially
- d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Quick Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort